Project Proposal Phase 1

Cryptonics Property Management System

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Glossary Terms

- **AML (Anti-Money Laundering):** Halting bad actors from hiding their illicit funds, these rules may step in. A business can ensure it isn't accidentally helping criminals.
- **API (Application Programming Interface):** Acting as a digital bridge, this tool can link completely different systems. Data may be shared quietly behind the scenes.
- **AWS (Amazon Web Services):** Housing the system off-site, this platform can host everything in the cloud. Expensive office servers may no longer be needed.
- **B2B (Business-to-Business):** Operating strictly with professional investors, this approach can bypass everyday retail buyers.
- **CRUD (Create, Read, Update, Delete):** Handling the four basic actions, users can manipulate database records. You may add a new tenant. You can also delete a file.
- **Cryptocurrency (Crypto):** Running on a decentralized network, this digital money can operate without a central bank.
- **Digital Escrow:** Locking money inside a neutral digital space, this tool can hold security deposits safely. Funds may automatically release once a co-host confirms no property damage.
- **Failover Router:** Stepping in during an outage, this backup internet device can keep the office online. It can prevent the cloud connection from dying if the main fiber drops.
- **Full-Duplex Bandwidth:** Allowing a computer to send and receive data at the exact same time, this network setup can stop signals from crashing into each other. It may feel like a two-way highway.
- **Hardwired Local Area Network (LAN):** Connecting office computers directly to the internet router, physical Ethernet cables may be utilized. A secure and lightning-fast connection can be created.
- **KYC (Know Your Customer):** Verifying who a person actually is, this legal process may check an ID or passport.
- **Legacy System:** Holding a company back from growing, these outdated software setups may still work.
- **MIS (Management Information System):** Crunching raw numbers into visual charts, this setup can help leaders make strategic choices.
- **POPIA (Protection of Personal Information Act):** Dictating exactly how private data may be collected, this strict South African privacy law can protect users.
- **ROI (Return on Investment):** Measuring the cash saved against the cash spent, this figure may justify the project budget.

- **Smart Contract:** Written in computer code, this digital agreement may act on its own. A deposit can be released instantly when a lease ends.
- **SWIFT:** Sending messages across international borders, this traditional global banking network may take days to transfer money.
- **Webhook:** Sending instant digital nudges, this snippet of code can tell the database exactly when a foreign payment clears.
- **Zero Wireless Latency:** Eliminating the tiny delays that naturally occur through Wi-Fi, this setup can rely on physical cables. Transactions may happen instantly.

1. INTRODUCTION

This strategic proposal introduces the Cryptonics Property Management System. Managing international real estate portfolios can create intense administrative headaches. Atlas Premier Properties may feel utterly suffocated by manual spreadsheets. Multi-day banking delays often frustrate foreign investors. To map out a viable escape route from these legacy bottlenecks, this document is presented. A secure, automated web portal can offer immediate relief. Outlining exact scopes and strict budgets, the upcoming pages break down exactly how this digital transformation may happen.

2. BACKGROUND

South Africa can often be seen as a sought-after destination for foreign property investors looking to invest in South Africa's real estate, particularly Cape Town, driven by a mix of geopolitical "push" factors in their home countries and attractive "pull" factors in South Africa. In this niched property management market for the elite sits **Atlas Premier Properties**, and they've been operational for over 2 decades, acting as stewards for wealthy owners from Eastern Europe and Turkey, Atlas has been handling every part of a property's life.

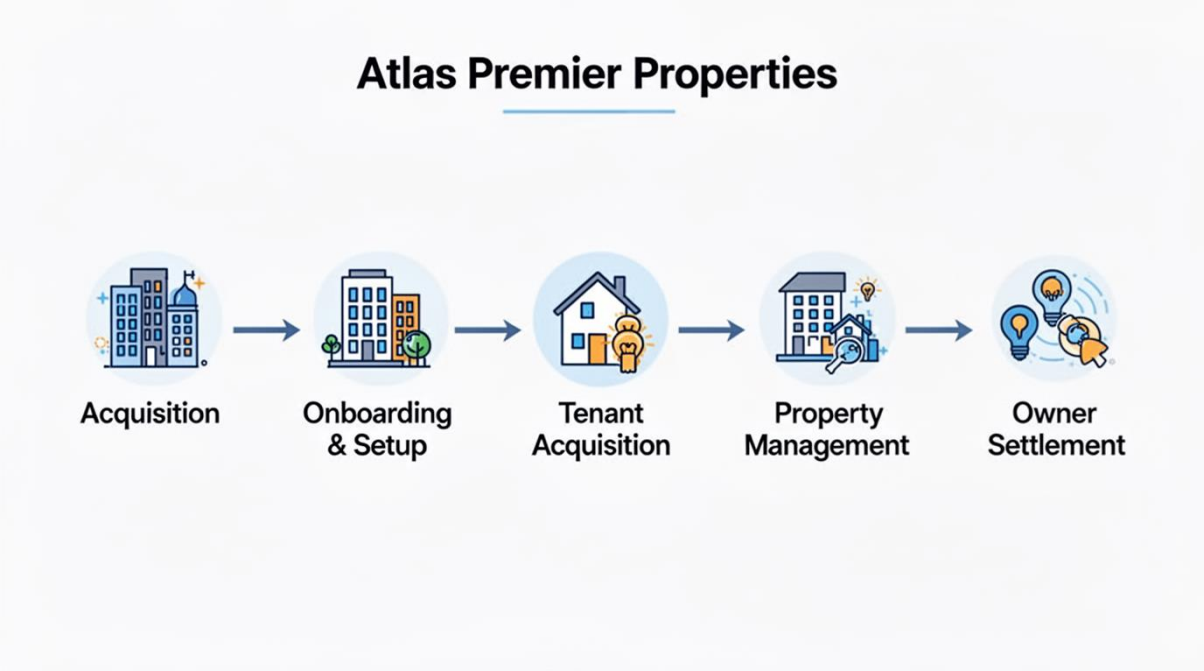

Figure 1: Property Life Cycle Diagram

As illustrated in Figure 1, the property life cycle outlines Atlas's core daily business processes. Atlas spends most of their time and resources finding international owners, orchestrating deals, and acquiring new properties to manage. Once a property is onboarded and prepared for the rental market, Atlas handles the manual vetting of prospective tenants and their credit documentation. Throughout the lease, they coordinate with local maintenance and cleaning staff to ensure the houses remain in good condition. Finally, Atlas collects the monthly rent, deducts their management fee, and settles the remaining profit with the owners.

The firm has been using a 15 year old legacy system to manage the property portfolio. Spreadsheets and emails have slowed their work down. The use of old banking systems like

SWIFT have led to delays in settlement and transfer times to and from the international property owners who use foreign banks.
This old setup we believe is not as competitive and has slowed down the process to secure new tenants. Due to the paperwork being so heavy, taking on new houses may feel impossible. Exploring the possibility of securing an advanced system is the only way for them to become profitable as they intend to reduce operational costs and improve ease of doing business including securing new tenants and managing existing tenants more efficiently.

Context:
Russian nationals suffer from market sanctions that restricted investors' access to traditional European and North American markets. Complete isolation,is another push factor, however, due to South Africa's non-aligned stance and membership in BRICS make it a conducive environment where their funds are less likely to be frozen or rejected.Ukrainian nationals seek stable, safe-haven destinations far from the conflict zone of the ongoing war for long term security, whilst Turkish nationals suffer from the high domestic inflation and a weakening home currency (Lira) push Turkish investors to move wealth into more stable emerging economies that are better-run metro with more reliable infrastructure and services. The Rand's exchange rate that offers a discount on luxury property compared to other global coastal hostposts.

3. PROBLEM STATEMENT

This section outlines the operational and financial problems currently hindering Atlas Premier Properties. It details seven key issues stemming from their outdated legacy systems. Including manual settlement errors, cross-border payment delays, fragmented data, and compliance risks, which collectively prevent the company from scaling.

1. **The Money Splitting Gap**: Between the property owner, the property manager, and the property management company, the way rental income is split is calculated manually. This can lead to mistakes and payouts are delayed due to these manual steps.
2. **Trouble with Sending Money Overseas**: For foreign property owners, the traditional banking systems cause a lot of distress. Delays/waiting times for money transfers by the Property Management Company to the Property Owner of between three to seven days is the norm. Also the fees for money transfers are very high and cut into the final profit.
3. **Slow Paperwork for New International Tenants:** Too many work hours are lost to checking international tenant papers. Passports and bank statements can take a long time to verify. The lease approval process may end up taking a while as a result.
4. **Fragmented information:** Data stored in spreadsheets and physical files are disconnected. Lease agreements sit in isolation. Generating a "Portfolio Health" dashboard view can feel impossible, as a result.
5. **Strategic Management Reports Are Missing**: Tools to generate clear reports in graphical formats are manual and take a long time to generate from their current system. Management find themselves making decisions based on incomplete information. Information on the revenue per property or how a real estate agent is performing is unattainable.
6. **Difficulty to Provide International Client Support**: For international tenants and owners, Atlas struggles to provide fast and immediate, real-time support. The time zone differences can complicate small questions regarding payments. Document issues last longer than they should.
7. **Lack of proper records and documents management:** Without a clear digital trail that can’t be changed, manually entering financial settlements/ details is risky. Staff members cannot be held responsible during the process. Year-end auditing can become a legal mess if the records are not kept well.

4. CONSTRAINTS

This section outlines the specific boundaries and limitations within which the Cryptonics system must be developed. It defines the strict financial budgets, deadlines, technology requirements, and the legal compliance measures necessary to protect the firm.

The following constraints present in the project:

- **Financing:** The project has a budget of **R705,000**.
- **Time:** The project must be designed and implemented by **21 October 2026**.
- **Technology:** The current hardware at Atlas needs to meet the minimum requirements for Windows 11 to ensure the new interface runs without any complications. The system and database will be written in Visual Studio. Microsoft Access must be used to build a relational Database to conform to the legalities and software licenses that the Property Management company already has.
- **Integration & Security:** Cryptonics will function as a supporting financial layer. A private blockchain cannot be built from scratch.
- **Legal & Regulatory (POPIA & AML):** Atlas is not licensed to store sensitive international passports. To avoid the legal trouble that comes with that, the system will be constrained to outsource from a third-party KYC (Know Your Customer) integration

Context:

The financial constraint of R705,000 was formulated through a cost-estimation model based on current South African market rates. The specifications were derived directly from the operational requirements of Atlas Premier Properties to ensure the system is stable, secure, and highly available:

- **Man-Hour Formulation:** Labor costs were calculated using standard local contracting rates for a development team of six part-time junior software developers over a 7-month period. It also factors in the proportional hourly rates for internal administrative staff and on-site technicians required for testing and implementation.
- **Hardware & Networking Specifications:** The budget specifically mandates Dell OptiPlex workstations hardwired via Ethernet into a 24-port Gigabit Network Switch. While Cryptonics is hosted on AWS, establishing a hardwired Local Area Network (LAN) for stationary office equipment guarantees full-duplex bandwidth, zero wireless

latency during financial API calls, and stronger physical security for transmitting sensitive KYC (Know Your Customer) data to the cloud.

- **Software & Licensing Formulation:** Software estimates include standard licensing fees for Microsoft products (Windows 11 Pro, Microsoft 365) to maintain compliance with Atlas's existing software ecosystem, combined with the estimated operational costs for AWS Cloud Web Hosting and SSL certificates.

5. SCOPE DEFINITION

This section defines the functional boundaries of the proposed system. It outlines the core database entities to be managed, the specific business workflows to be automated (such as tenant vetting and crypto split-payments), and the strategic graphical reports that will be generated for management.

The system will act as a centralized, secure web portal serving as the heart of the operation. It will oversee long-term property leases and manage co-hosts. Automated financial settlement splits will happen in the system, as well.

Entity Maintenance (CRUD):

The system must include functionality for the following:

1. **Maintenance of Properties**
2. **Maintenance of Property Owners**
3. **Maintenance of Tenants**
4. **Maintenance of Property Manager**
5. **Maintenance of Lease Agreements**
6. **Maintenance of Settlements**
7. **Extensive Help functionality**

Business Processes Automated:

- Tenant vetting process based on API (Application Programming Interface) verification statuses and income criteria.
- Generation of Cryptocurrency payment invoice links for approved tenants. Cryptocurrency enables instant, low cost cross border settlements and is legally compliant when facilitated through established API payment gateways that interface with local and international banks.
- Split-Payment settlement logic calculation of the owner's share and the management fee after payment has been made.

Reports for Management:

The system will provide real-time strategic reporting in graphical formats:

- **Co-Host Performance Metrics:** A bar chart showing who is renting out the most properties. It will compare days rented against total days (occupancy rate). Helping management to identify overperforming and underperforming property managers.

- **Financial Friction Report:** For every crypto payment received, the system will run a quick test. It will compare the low crypto fees against the high bank fees. A simple line graph will show how much money is saved over a whole year. Showing the Return-On-Investment of using the new system.

Context:

CRUD, stands for Create, Read, Update, and Delete, this concept outlines the four basic ways a user may interact with a digital file.

To manage operations, each entity serves a distinct purpose:

- **Maintenance of Properties:** Staff can track physical houses and instantly update vacancy statuses.
- **Maintenance of Property Owners:** Investor profiles may be created, safely storing sensitive crypto wallet details.
- **Maintenance of Tenants:** Crucial vetting details and verification statuses can be securely maintained.
- **Maintenance of Property Managers:** Local property managers may be overseen, allowing management to track their specific commission rates.
- **Maintenance of Lease Agreements:** Rental boundaries can be clearly managed. Staff may log start dates and escrow terms here.
- **Maintenance of Settlements (Create & Read Only):** Acting as an unchangeable vault, this area exclusively logs crypto payouts. Update and delete functions are permanently disabled to forge a digital trail that no one can change.
- **Extensive Help Functionality:** Built-in digital guides may assist users navigating the new interface.

6. GOALS OF THE PROJECT

This section outlines the primary objectives of the proposed system. Focusing on optimizing operational processes through automated financial settlements, streamlining international communications via cryptocurrency integration, and enhancing strategic decision-making through real-time analytical dashboards.

The following goals were identified:

- The system aims to improve business processes by automating the split payment calculation and distribution of rental income to all stakeholders.
- The system aims to improve business communications by mediating the cross-border payment gap through a crypto currency integration, to allow international owners to receive funds almost immediately without various email message chains.
- The system aims to improve business knowledge by giving the property Management Company access to real time information reports and graphical dashboards that show the health and financial savings of assets.

7. OPPORTUNITIES TO IMPROVE

This section identifies value adding features that elevate the system beyond basic property management. These suggested enhancements focus on improving financial security, automating tax compliance, reducing currency risks, and breaking down international language barriers.

The following opportunities to improve the system beyond the scope of definition were identified:

- To improve financial security by incorporating a Crypto Smart Contract to build trust with international tenants and eliminate the risk of funds being withheld unfairly.
- Creating an automation to generate one click spreadsheet tax reports for our clients, in order to show them their net rental income with the management fees already deducted.
- To improve revenue protection for international owners by creating an automation that tracks the exchange rate between the cryptocurrency and the South African Rand to protect profit margins and protect against currency volatility.
- To improve customer support and the speed of the tenant onboarding process by combining Amazon Lex AI chatbot into the web portal to solve global customer disputes anytime, regardless of the time zone and language barrier problems.

8. SCHEDULE

This section outlines the proposed Systems Development Life Cycle (SDLC) timeline for the project. It details the key milestones and deliverables, tracking the system's progression from initial scope and problem analysis in March 2026 through to final construction and testing in September 2026.

Project Steps	Steps Implemented	Proposed Date
Scope definition	The overall scope to be documented in the project proposal.	6/3/2026
Problem analysis	Document: ● The problems with the current system ● The solutions to the troubled system ● The opportunities that the new system can make to help improve the current system ● The proposed schedule to be set	6/3/2026
Requirements analysis	● Graphical user interface requirements ● Functional data, process and interface requirements for system ● Non-functional requirements for system	27/4/2026
Logical design	Specifications and the logical system modules	27/4/2026

Decision analysis	• Analyse every candidate solution in terms of technical, operational, economic, schedule and risk feasibility • Decide on the best candidate solution	08/5/2026
Physical design & integration	• Physical business design, software design, and physical user and system interface specifications • Design prototype • Physical design modules • Redesigned business processes	4/7/2026
Construction & testing	• Create Database • Construct software program • Construct User Interface • Test user interface for usability and stability	21/10/2026

9. BUDGET

This section outlines the total estimated financial investment required to develop and implement the system over a 7-month lifecycle. The expenditures are systematically categorized into human resources (man-hours), physical hardware infrastructure, and essential software licensing, culminating in a total projected budget of R 698,880.00.

HOURS

Item	Units / Duration	Rate (R)	Calculation	Cost (R)
Admin Clerk (20% allocation)	**20% × 1 person × 8 hrs × 217 days**	**R250/hr**	**0.20 × (1 × 8 × 250 × 217)**	**R 86,800.00**
Technicians (2 people, 2 days/week for 7 months)	**2 × 56 days × 8 hrs**	**R130/hr**	**2 × 56 × 8 × 130**	**R 116,480.00**
Consultants (6 junior developers part-time)	**6 × 3 hrs × 217 days**	**R100/hr**	**6 × 3 × 217 × 100**	**R 390,600.00**
Total Man-hour Cost				**R 593,880.00**

HARDWARE

Item	Quantity	Unit Price (R)	Calculation	Cost (R)
Dell OptiPlex All-in-One	5	R 15,000.00	5 × 15,000	R 75,000.00
Dell Keyboard and Mouse	5	R 400.00	5 × 400	R 2,000.00
Network Switch (24-Port)	1	R 1,500.00	1 × 1,500	R 1,500.00
Enterprise Wi-Fi Router / Firewall	1	R 3,500.00	1 × 3,500	R 3,500.00
Total Hardware Cost				R 82,000.00

SOFTWARE

Item	Quantity / Duration	Unit Price (R)	Calculation	Cost (R)
Windows 11 Pro Licence	5	R 1,300.00	5 × 1,300	R 6,500.00
Microsoft 365 Licence	5	R 900.00	5 × 900	R 4,500.00
AWS Cloud Web Hosting	7 months	R 1,500/month	7 × 1,500	R 10,500.00
Domain Name + SSL Certificate	1	R 1,500.00	1 × 1,500	R 1,500.00
Total Software Cost				R 23,000.00

PROJECT COST

Cost Category	Amount (R)
Total Man-hour Cost	R 593,880.00
Total Hardware Cost	R 82,000.00
Total Software Cost	R 23,000.00
TOTAL PROJECT COST	**R 698,880.00**

10. INTERNAL RESOURCES

The suggested internal resources for the project are:

- **System Owner (Sponsor)**: Mr. J Browns
- **System Designer**: Ms S Mabaso
- **Project Manager**: Ms S Gawu
- **System Builder**: Ms M MABOMBO
- **System Developer:** Ms Z Nghondzweni
- **System User**: Mr Z Hansa
- **System Analyst:** Mr TJ Mabuza

The following system users are proposed and should be available for 20% of their time for 8 months:

- **Administrative Clerk (Vetting):** Lerato Ndlovu
- **2 Technicians called out twice a week**: John Bells and Chris Nolan

11. CONCLUSION

In conclusion, the Cryptonics system can totally rewire how Atlas Premier Properties functions daily. Stripping away the risk of human error, automated split settlements may instantly speed up global transactions. Staff members can stop chasing paperwork, this will allow them to start focusing on acquiring new international clients. Proving that the budget and timeline are realistic, this proposal outlines a highly achievable project. Upgrading this core technology may do much more than just fix old spreadsheets. It can propel the entire company into a new era of global real estate management.

11. APPENDIX

Appendix A: Detailed Project Budget Breakdown.

Role / Responsibility	Assigned person	Availability	Units / Duration (used for costing)	Rate (R)	Calculation	Cost (R)	Notes
System Owner (Sponsor)	Mr J Browns	as required	N/A	N/A	N/A	N/A	Sponsor / decision authority
System Designer — (part-time developer)	Ms S Mabaso	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs
Project Manager — (part-time developer)	Ms S Gawu	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs

System Builder — (part-time developer)	Ms M MABOMBO	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs
System Developer — (part-time developer)	Ms Z Nghondzweni	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs
System User — (part-time developer)	Mr Z Hansa	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs
System Analyst — (part-time developer)	Mr TJ Mabuza	part-time dev	3 hrs/day × 217 days	100 / hr	3 × 217 × 100	R 65,100.00	Counted as one of the 6 part-time devs
Subtotal — 6 part-time developers			6 × (3 × 217 × 100)		6 × 3 × 217 × 100	**R 390,600.00**	6 × R 65,100

Administrative Clerk (vetting)	Lerato Ndlovu	**20% availability**	20% × 1 person × 8 hrs × 217 days	250 / hr	0.20 × (1 × 8 × 250 × 217)	**R 86,800.00**	Included in man-hour cost
Technician (on-call)	John Bells	Called out 2×/week (full day)	1 × 56 days × 8 hrs	130 / hr	1 × 56 × 8 × 130	R 58,240.00	Half of technicians subtotal
Technician (on-call)	Chris Nolan	Called out 2×/week (full day)	1 × 56 days × 8 hrs	130 / hr	1 × 56 × 8 × 130	R 58,240.00	Half of technicians subtotal
Subtotal — Technicians (2)			2 × (56 × 8 hrs)		2 × 56 × 8 × 130	**R 116,480.00**	2 × R 58,240
Total Man-hour Cost (Admin + Techs + 6 devs)						**R 593,880.00**	
Dell OptiPlex All-in-One	Dell Technologies	5 units	5 × unit	R15,000.00	5 × 15,000.00	R 75,000.00	Hardware

Keyboard & mouse (set)	EveTech	5 sets	5 × unit	R400.00	5 × 400.00	R 2,000.00	Hardware
Network switch (24-port)	EveTech	1	1 × unit	R1,500.00	1 × 1,500.00	R 1,500.00	Hardware
Enterprise Wi-Fi router / firewall	EveTech	1	1 × unit	R3,500.00	1 × 3,500.00	R 3,500.00	Hardware
Subtotal — Hardware						**R 82,000.00**	
Windows 11 Pro licences	Microsoft Corporation	5 licences	5 × unit	R1,300.00	5 × 1,300.00	R 6,500.00	Software
Microsoft 365 licences	Microsoft Corporation	5 licences	5 × unit	R900.00	5 × 900.00	R 4,500.00	Software
AWS cloud web hosting (7 months)	Amazon Web Services	7 months	7 × monthly fee	R1,500.00 / month	7 × 1,500.00	R 10,500.00	Hosting
Domain + SSL certificate	Namecheap.com	1	1 × unit	R1,500.00	1 × 1,500.00	R 1,500.00	Software / services

Subtotal — Software & services						**R 23,000.00**	
TOTAL PROJECT COST						**R 698,880.00**	Man-hour + Hardware + Software

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